

Study on the Impact of Detector Replacement on Accuracy and Coding Efficiency in Video Coding for Machines

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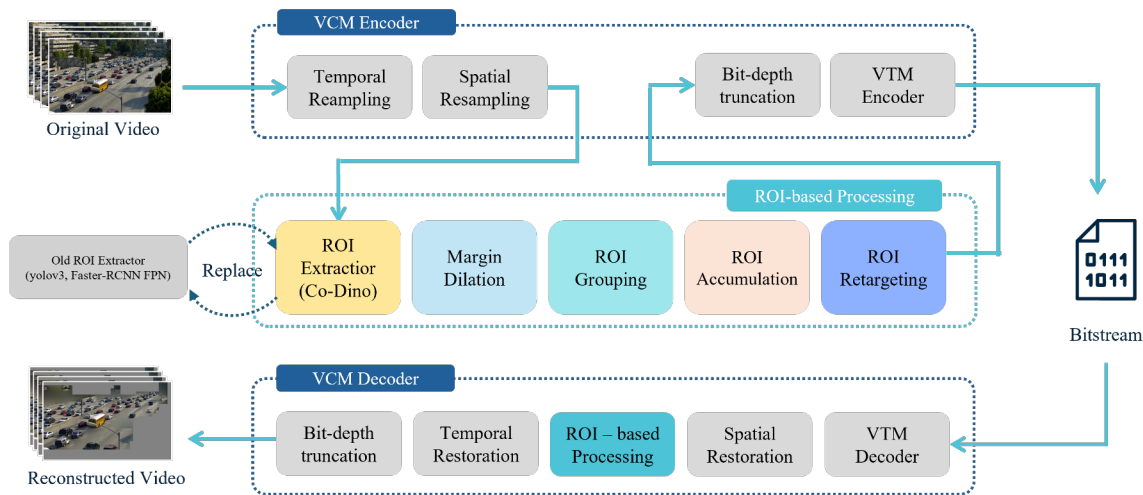


Figure 1: Overview of the framework

ABSTRACT

Recently, the video compression industry has increasingly focused on developing compression approaches tailored to machine vision applications. Enhancing detection accuracy is essential to preserve critical information. We show that replacing the legacy ROI-generation detector in the MPEG Video Coding for Machines reference software (VCM-RS) with a state-of-the-art detection model delivers better preservation of object information and improved reliability compared with the existing baseline. In the current implementation of the MPEG VCM-RS, the ROI module uses older detectors that often miss small or occluded objects. The study measures end-to-end impact on the random access scenario. Average detection accuracy increases from 47.84 to 51.16 mAP and for certain sequences, coding efficiency also improves. This paper highlights the need for broader testing of recent detection models within VCM-RS.

Index Terms: Video coding for machines, ROI-based coding, Object detection, Video Coding

1 INTRODUCTION

Recent studies have increasingly focused on allocating bits more efficiently by prioritizing machine vision over human perception [3, 5, 9, 10, 4]. Video Coding for Machines (VCM) [1] project, which is led by MPEG, compresses video frames (or features) for machine vision tasks rather than human viewing. VCM-reference

software (VCM-RS) includes temporal, spatial resampling, bit depth truncation, and region of interest (ROI)-based coding. In ROI coding, detector boxes are expanded and merged into a union mask. Pixels inside the mask keep their original RGB values, while pixels outside are replaced by a uniform mid-gray background.

In the current implementation of the MPEG Video Coding for Machines reference software (VCM-RS), the ROI module uses Faster R-CNN X101-FPN [8, 6] and YOLOv3 [7] for object detection. These detectors were introduced several years ago and may fail to detect small, occluded, or low-contrast objects in challenging scenes. When such objects are missed, they fall outside the mask and are suppressed, as shown in Figure 2. This can lead to the loss of critical information and a decline in coding efficiency.

However, in MPEG VCM, there is still a lack of research on preprocessing with diverse recent detection models, and the framework continues to rely on a fixed set of detectors. In this study, we replace the ROI detector in MPEG VCM-RS with a more recent detection model and subsequently evaluate detection accuracy and coding efficiency across each sequence.

2 EXPERIMENT

As shown in Figure 1, we evaluate a detector replacement in the VCM-RS ROI module by replacing the baseline Faster R-CNN X101-FPN with a recent detection model (Co-DINO ViT-L [11]). The experiment is based on VCM-RS version 1.0 and uses the SFU-HW-Objects-v3.2 dataset [2] as the test dataset. The study follows the Random Access Inner (SFU_RA_Inner) scenario and reports results over the official class partitions A–D. Metrics include mean Average Precision (mAP) for detection and BD-rate and BD-mAP for end-to-end coding performance. The procedure runs the full preprocessing and encoding pipeline with the detector as the only changed element and holds all remaining settings fixed.

Table 1 reports detection accuracy. Average mAP rises from

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Figure 2: Comparison of (a) the original frame and (b) the ROI-masked result.



Figure 3: Comparison of (a) the ROI-masked result using Faster R-CNN X101-FPN and (b) Co-DINO.

Table 1: Per-class mAP performance comparison on SFU_RA.Inner scenario

	Anchor Avg mAP	Co-DINO ViT-L Avg mAP
Class A	53.95	61.39
Class B	46.73	47.40
Class C	48.61	49.30
Class D	42.08	42.80
Average	47.84	51.16

Table 2: Evaluation result comparison between anchor and Co-DINO for ROI generation on SFU_RA.Inner scenario

	End-to-End BD-Rate [%]	End-to-End BD-mAP
Class A	-0.32	0.14
Class B	22.99	-2.20
Class C	-7.75	0.35
Class D	-7.26	0.56
Average	0.72	-0.23

47.84 to 51.16, with improvements observed across all four classes. The gain is most notable in Class A, where mAP increases from 53.95 to 61.39.

Table 2 shows the end-to-end BD-rate and BD-mAP results on the SFU_RA.Inner scenario. On average, BD-rate increases by 0.72%, while BD-mAP decreases by 0.23. Class-level outcomes are not uniform: Class B shows a clear increase in bitrate cost,

while Classes C and D achieve bitrate savings.

Figure 3 illustrates that replacing the detector leads to improved coverage of vehicles in the scene, reducing missed detections and preventing relevant objects from being suppressed during the pre-processing stage.

Overall, the results show that newer detectors can improve detection accuracy, thereby reducing the loss of critical object information. However, the impact on bitrate varies across sequences. In particular, the large increase in BD-rate for Class B indicates that a model change does not guarantee overall benefit in every case. The evidence suggests that updating the ROI detector may help preserve objects more effectively, but the trade-off with coding efficiency requires further study before a firm conclusion can be drawn.

3 CONCLUSION

This study examined the impact of replacing the legacy detectors in the MPEG VCM-RS ROI module with a more recent detection model. Experimental results demonstrated consistent improvements in detection accuracy and better preservation of critical object information, underscoring the necessity of updating ROI-generation components in VCM pipelines. Although the effect on coding efficiency was not uniform across all sequences, the findings indicate that detector replacement exerts a notable influence on end-to-end performance in video coding for machines. This suggests that integrating more recent models can provide a meaningful contribution to the robustness and reliability of machine-oriented video compression. Future work will extend the evaluation to a broader set of detectors and datasets, and will further explore adap-

tive ROI generation strategies that jointly optimize recognition accuracy, coding efficiency, and task-specific performance across diverse application scenarios.

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